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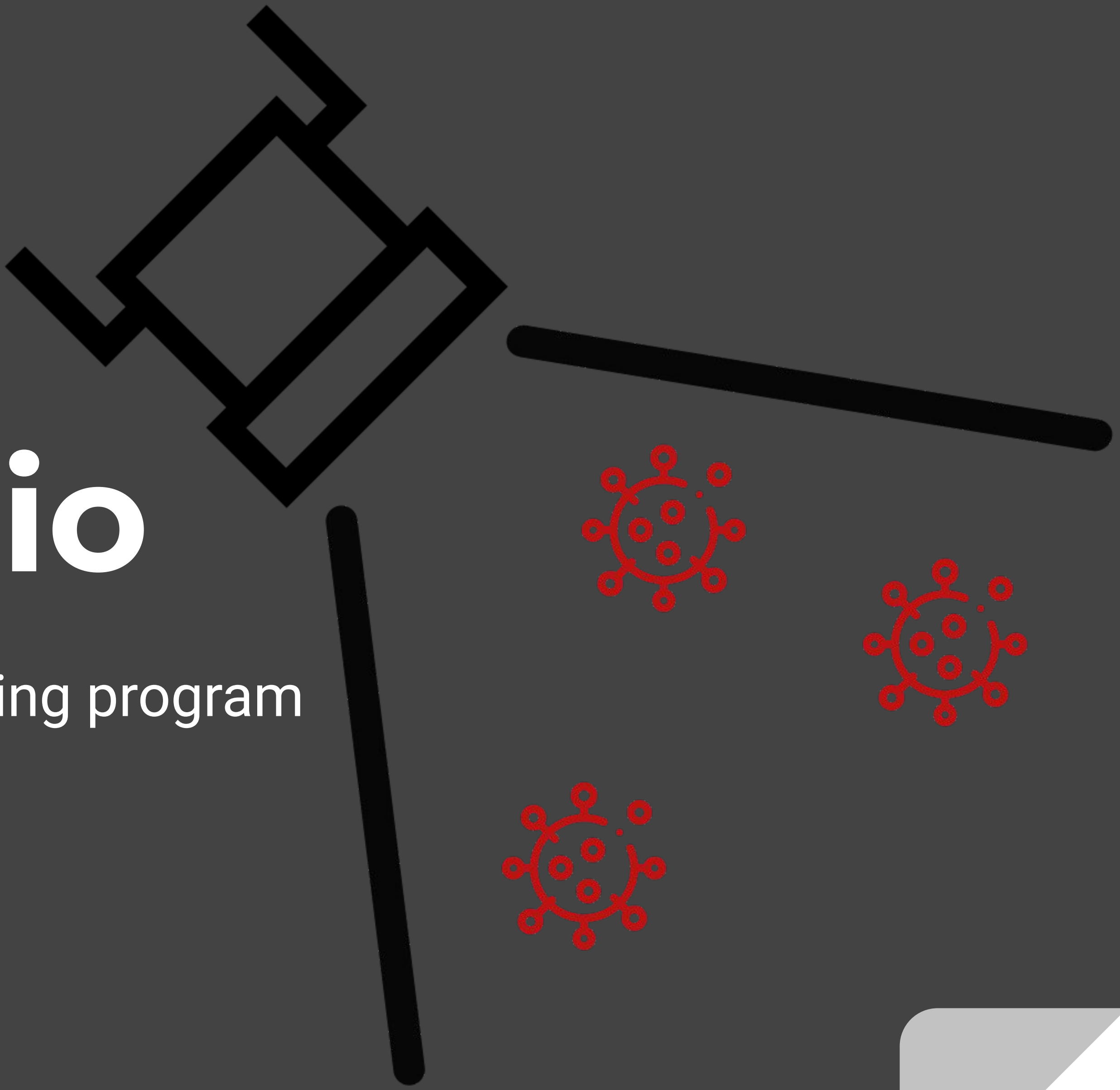
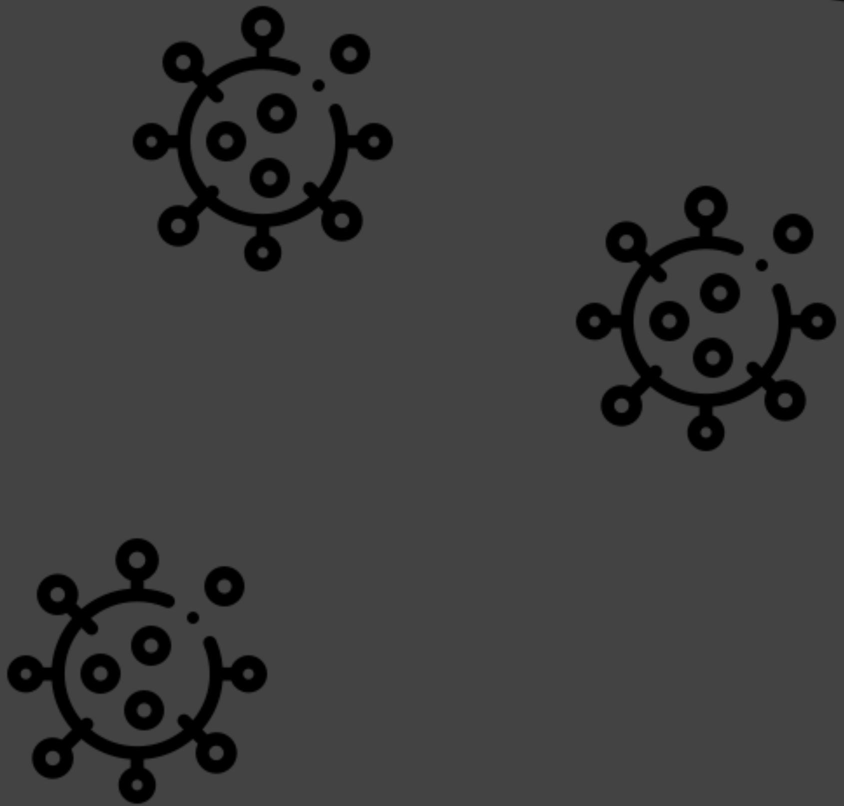


A nonprofit-based COVID-19 mass screening program

Focus on Foundations

in collaboration with
OpenCovidScreen

Mission statement: End COVID-19 Crisis with scalable, open-sourced mass screening



Who we are

Contributors / Advisors / Collaborators

- **Randy True** – *Founder Focus on Foundations, 501(c)3 STEM Education Nonprofit; Founder True Materials Inc; QB3 Garage Alum!; former VP of R&D at Affymetrix; Stanford Physics*
- **Theresa Ling** – *UX Design, former UX Lead at Uber, Lead Product Designer at New Relic*
- **Zarak Khurshid** – *Asymmetry Capital, former MDx Analyst at Wedbush*
- **Tim Lugo** – *Biotechnology Group at William Blair*
- **Brian Rabe** – *Harvard Medical School graduate student; COVID-19 diagnostic assay developer*
- **Connie Cepko** – *Professor of Genetics and Opthamology at Harvard Medical School; HHMI*
- **Jeff Huber** – *Founder OpenCOVIDScreen; Founding CEO & Vice Chairman of GRAIL; former Google SVP; board member at Parker Institute*
- **Cliff Wang** – *Lead OpenCOVIDScreen; former Illumina Infectious Disease Diagnostics Scientist; former Assistant Professor of Chemical Engineering at Stanford*
- **Laura Buxton** – *Biochemistry Teacher, Menlo High School*
- **Sonya Lebedeva** – *intern, Menlo High alum, incoming Berkeley Freshman*

Executive Summary

The United States is experiencing a severe public health crisis and we lack a key tool in this fight against COVID-19.

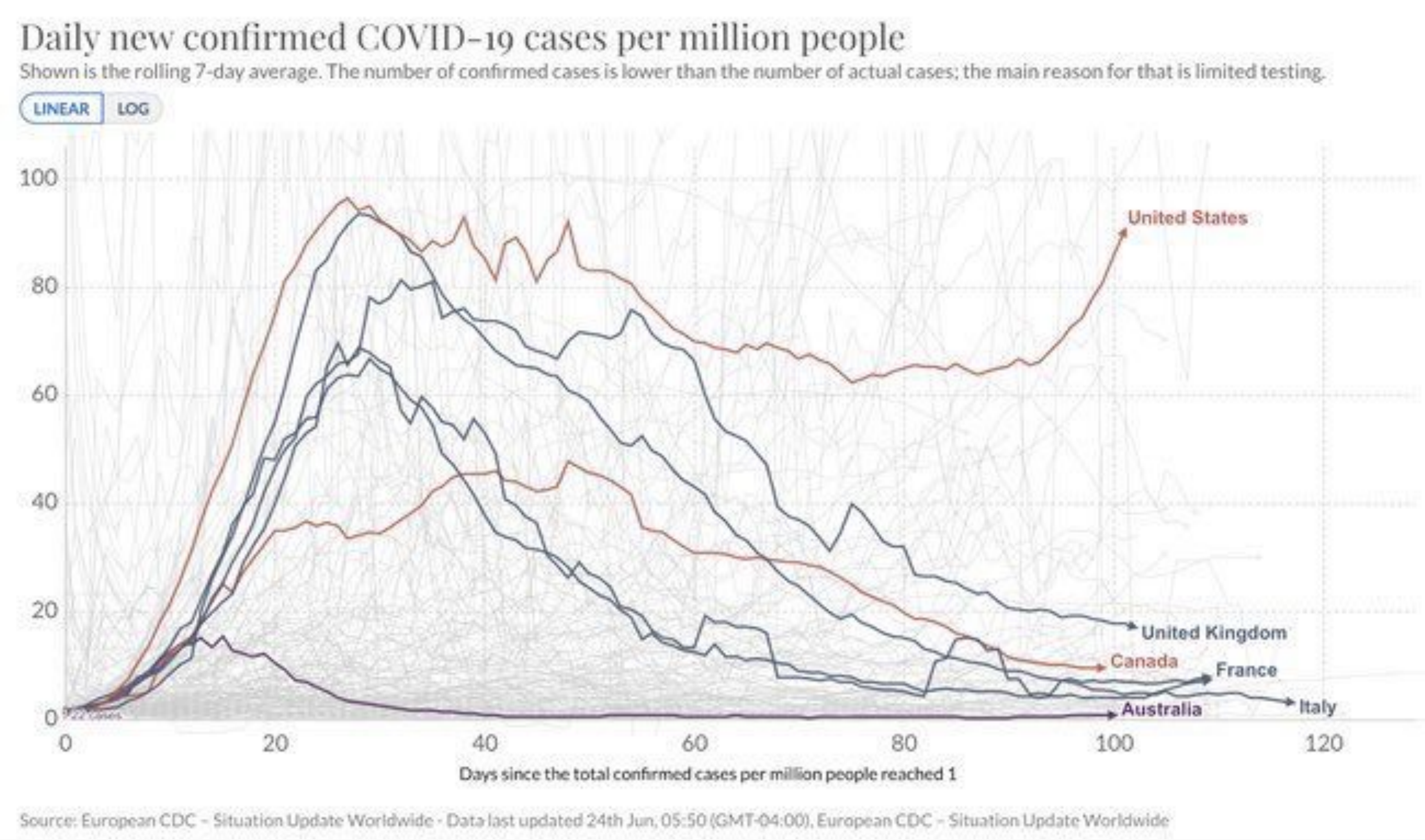
To address this urgent need we are pulling out all the stops to build a Nonprofit, Open, Integrated Mass Screening Program:

- **Widespread Outreach - raise awareness and drive adoption; key avenue through schools.**
- **Registration App - quickly, smoothly request screening as indiv, family, group, school, company, or town.**
- **Sample Collection Site - optimized coordination of participants at fixed and flex sites, efficient staffing, safe collection of immediately pooled samples.**
- **Detection Assay - ultra low cost, sensitive, robust; key LAMP amplification reagent already FDA approved**
- **CLIA Lab Network - scalable business model to leverage existing CLIA labs w low footprint, low cost, low complexity equipment set need; utilize existing lab staff or newly trained techs.**
- **Centralized Reagent Production - by nonprofit consortium and distributed to partner CLIA labs on transparent prioritized basis.**

The Problem

The U.S. Government and Commercial Diagnostics Industry are continuing to fail at delivering sufficient COVID-19 Testing.

The U.S. is Struggling



Systemic Failures

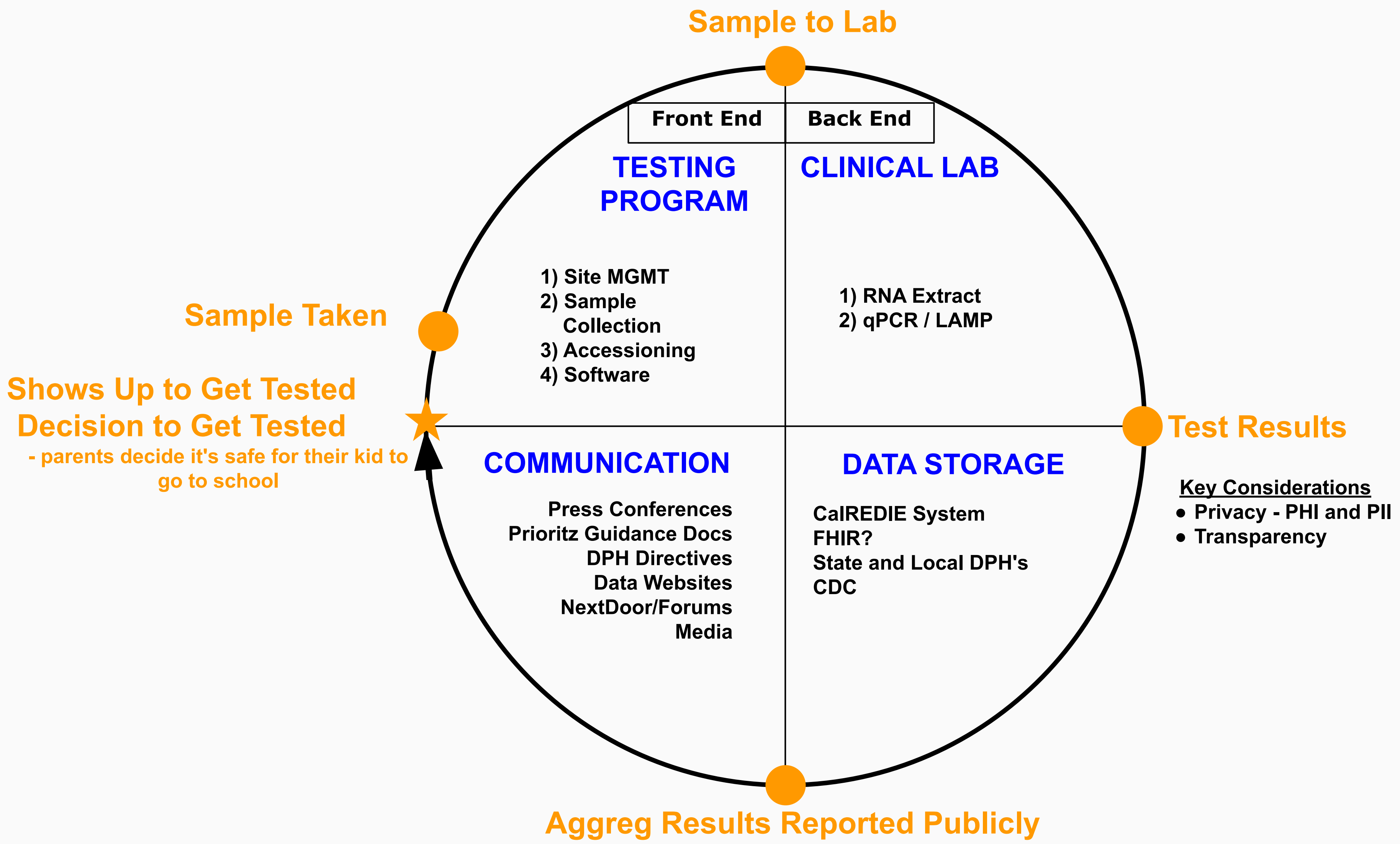
Incentives are simply not aligned to facilitate fast progress on testing.

Lack of federal leadership had led to each state scrambling to solve the problems themselves, while they simultaneously face massive budget deficits.

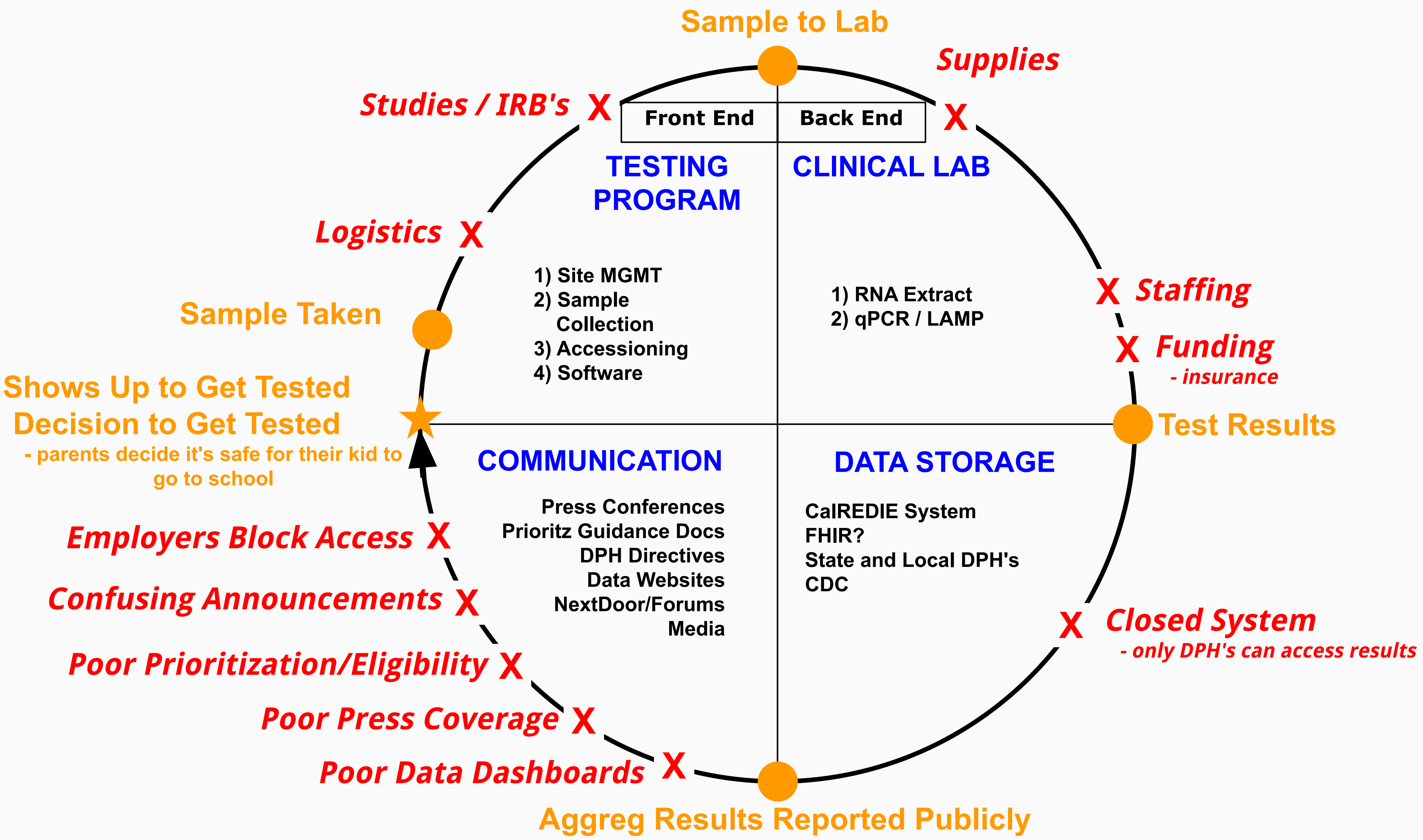
Regulations from FDA and CDC have been tragically slow to adapt to the scale of the crisis.

NIH/BARDA RadX has yet to deliver scalable solutions despite $1.4B in funding and mandate for impact by the end of summer.

COVID-19 Dx Testing Cycle



COVID-19 Dx Testing Cycle Problems



Public Schools in Jeopardy

Top Universities and Elite Private High Schools are putting in place testing capability now. Some are building and some are buying.

Public K-12 schools and many Community Colleges have no plan for testing and no expectation of significant help.

Largest college system in the country, Cal States, have announced they will be online only in the fall.

Having our public schools closed or reopening unsafely will be a national tragedy.

The Solution

A nonprofit driven, largely non-commercial effort that integrates all aspects of mass screening in an open-sourced, scalable program to achieve universal coverage and end the COVID-19 crisis.

Nonprofits / Philanthropy Needed

Gaps between IVD companies, CLIA labs, and Testing Program Providers can be filled by a non-commercial consortium of well connected, well funded nonprofits.

Focus on the public health, along with total transparency, is needed to garner trust and encourage participation.

There is strong synergy with our public education system because opening schools safely requires mass screening.

Return on investment for funding this nonprofit effort will be immense.

This effort establishes a precedent for future, possibly more deadly pandemics.

Tide Turning to Pooling

FDA U.S. FOOD & DRUG ADMINISTRATION

Coronavirus (COVID-19) Update: Facilitating Diagnostic Test Availability for Asymptomatic Testing and Sample Pooling

For Immediate Release: June 16, 2020

Statement From: Director - CDRH Offices: Office of the Center Director
Dr. Jeffrey E. Shuren MD, JD

Related Information

- Molecular Diagnostic Template for Manufacturers (updated June 16, 2020)
- Molecular Diagnostic Template for Laboratories (updated June 16, 2020)
- FAQs on Testing for SARS-CoV-2

e) Sample pooling with a new test (not previously authorized)

The recommendations below reflect FDA's current thinking. The study design and requirements may change as additional information becomes available regarding viral titer dynamics and transmission rates.

FDA stated:

- **high sensitivity is regarded for screening/pooled assays**
- **EUA submission is required for pooled testing**

What's Popular? In Molecular Diagnostics

1 At ASM Microbe, White House's Deborah Birx Urges COVID-19 Test Pooling and Community Outreach

2 Improved Pool Testing Methods Could Enable COVID-19 Screening as Pandemic Continues

June 23: Dr. Anthony Fauci, CDC Director Dr. Robert Redfield, Admiral Giroir, and other top health officials testified before the House Energy and Commerce Committee

Dr. Redfield: (43:34)

we're looking at ways that can really substantially enhance testing by potentially pooling samples. So right now, as Giroir said, we're doing 500,000, 600,000 tests a day. If we can pool samples five to one, that would bring it to three million tests a day. So we're continuing to try to enhance testing. It's a critical underpinning of our response.

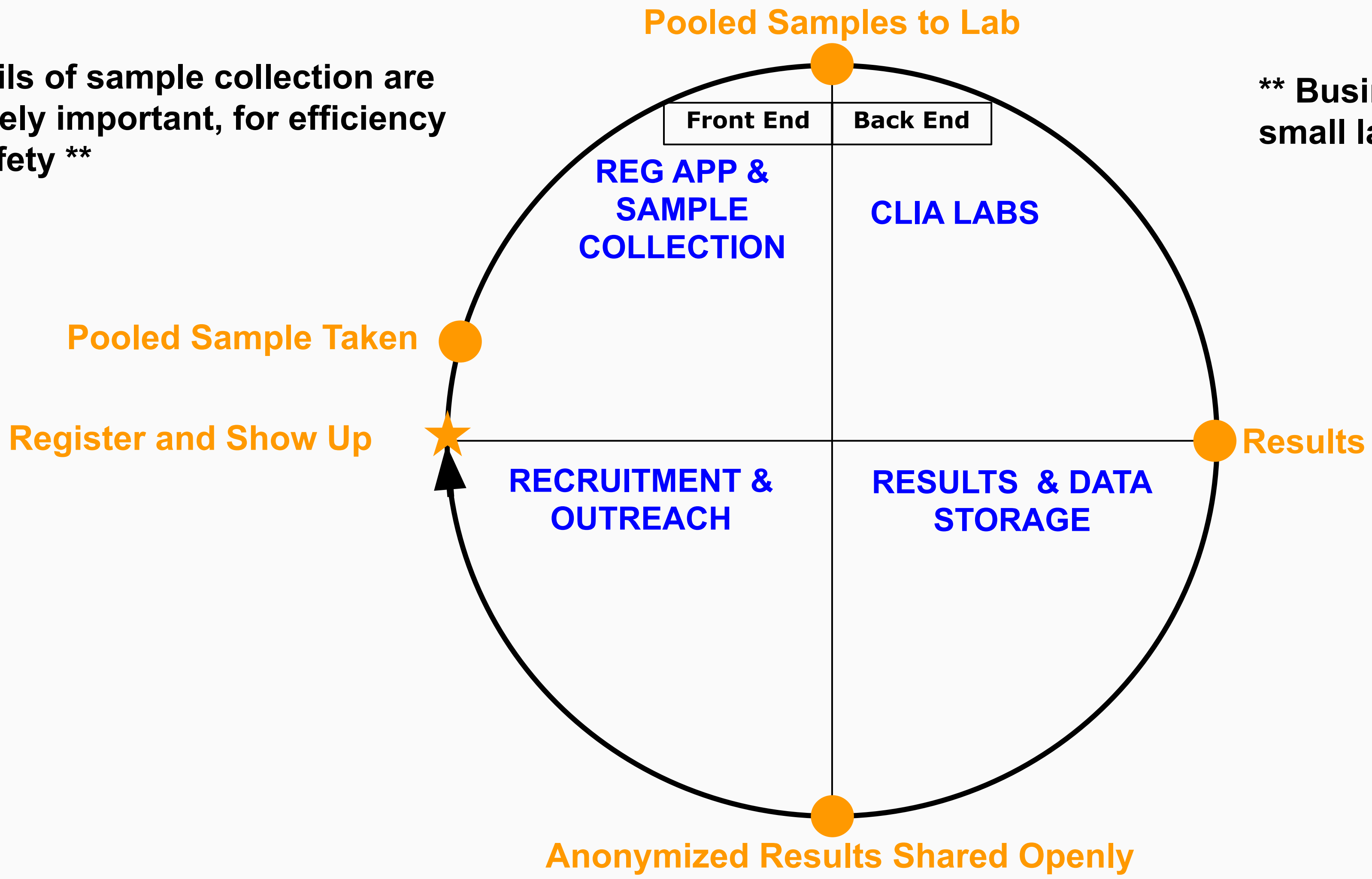
Admiral Giroir: (02:18:35)

Advances in technology, and that's not even including pooling. I do say pooling because the FDA just put up standards for validating pooling. So we would expect based on preliminary data we have, that on many tests, we can at least pool five to one and maybe up to ten to one. So when you do that math that I think will be validated very quickly by academic institutions and by large organizations, that

Mass Screening - 360° Integration

** Details of sample collection are extremely important, for efficiency and safety **

** Business model that brings in small labs is crucial **



Open Source and Universal

All protocols and procedures will be shared, in various media formats and at various levels of expertise.

Explicit guides will be produced, in vein of the IGI Doudna Lab's "Blueprint for a Pop-Up SARS-CoV-2 Testing Lab".

Systems and staffing will be put in place to support scaling.

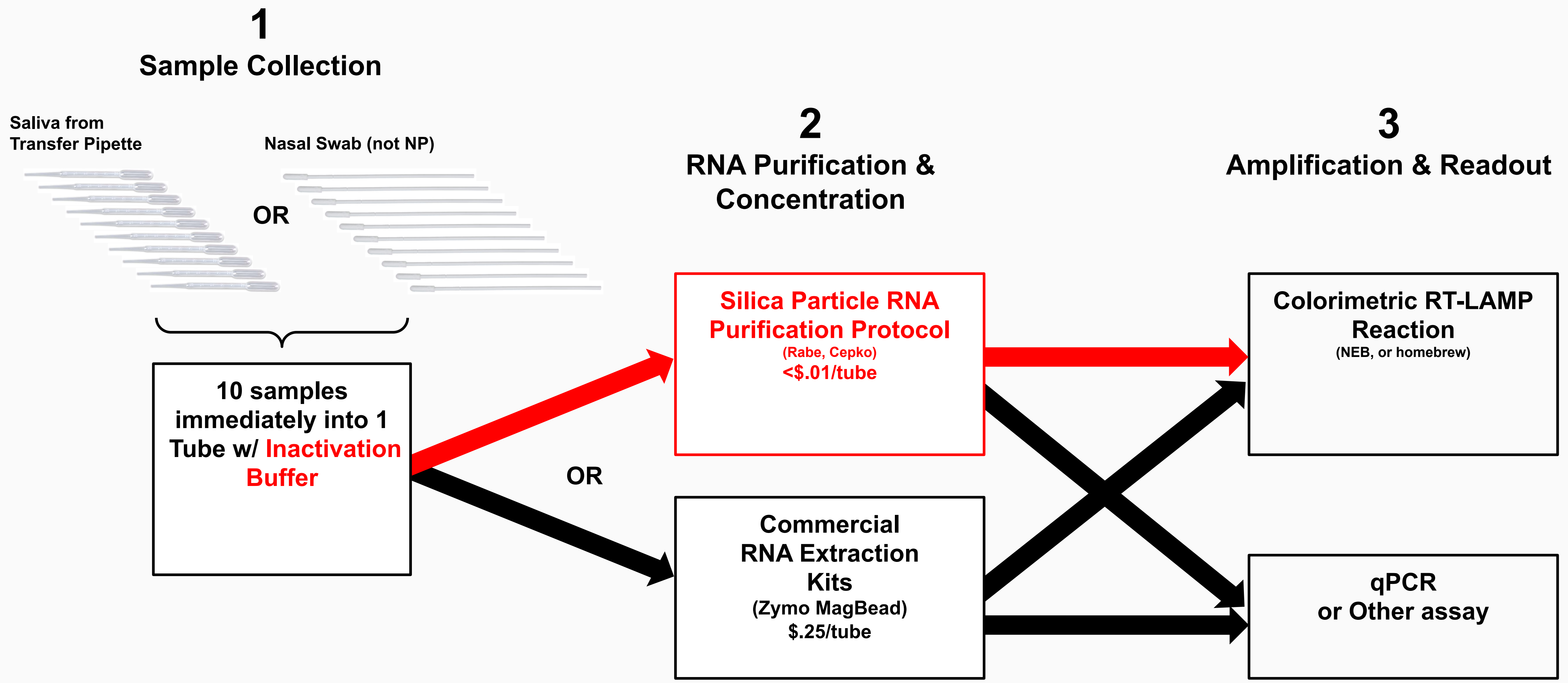
Goal is to enable any group sufficiently motivated group to bring screening to their community, and then both expand their coverage and help others do the same.

Further goal is for everyone to participate in COVID-19 screening.

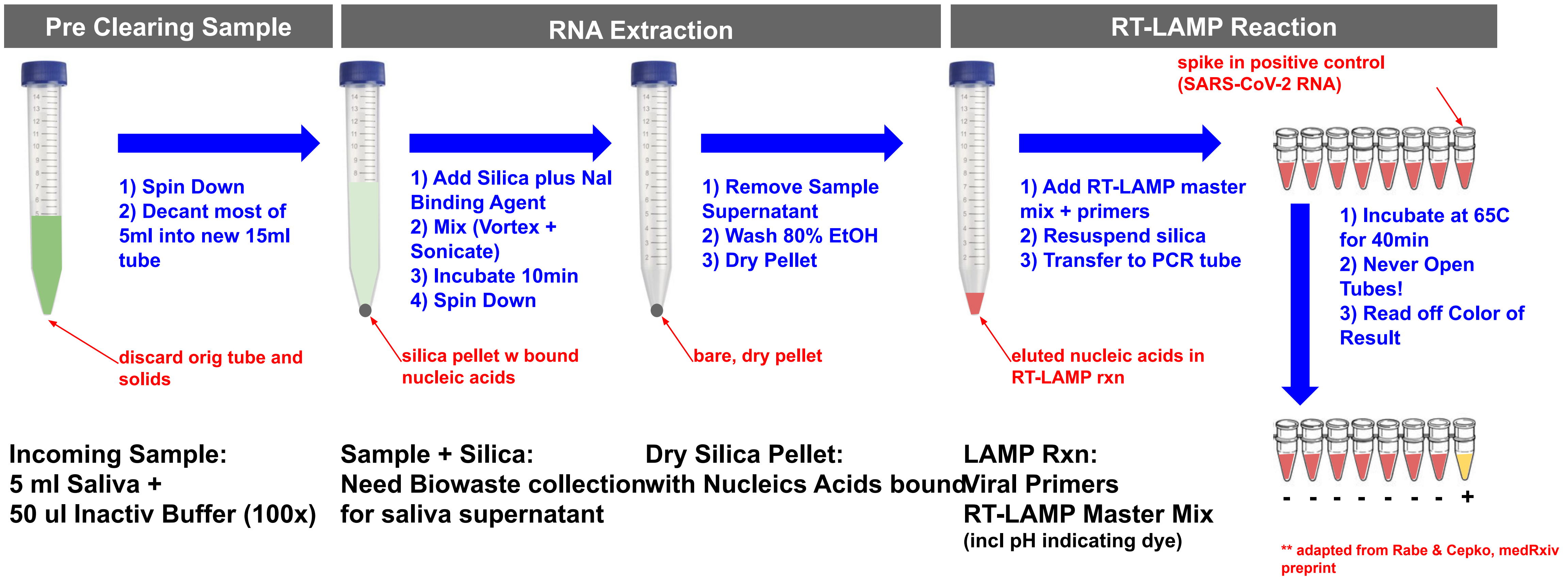
The Details

Cheap, fast, sensitive, robust assay chemistry with wrap-around program components of software, site/sample collection, logistics, CLIA lab network, reagent+supply distribution, and educational outreach to drive adoption.

Fast, cheap Inactivation & Purification



v1.0 Assay



** One 15ml Falcon tube w 10 samples ends up in a 200ul PCR tube **

Rabe-Cepko Streamlined RT-LAMP

SARS-CoV-2 Detection Using an Isothermal Amplification Reaction and a Rapid, Inexpensive Protocol for Sample Inactivation and Purification

Brian A Rabe* and Constance Cepko*

*Harvard University Medical School

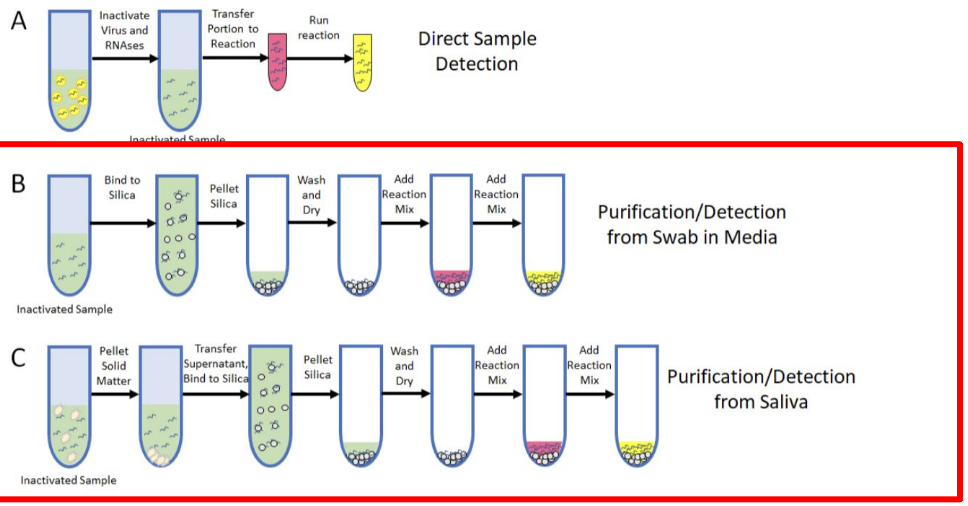


Figure 8. A Simple and Rapid Sample Inactivation and Purification Scheme

Clinical assessment and validation of a rapid and sensitive SARS-CoV-2 test using reverse-transcription loop-mediated isothermal amplification

Melis N Anahtar, Graham EG McGrath, Brian A Rabe, Nathan A Tanner, Benjamin A White, Jochen K.M. Lennerz, John A. Branda, Constance L. Cepko, Eric S Rosenberg

doi: https://doi.org/10.1101/2020.05.12.20095638

Visual Readout

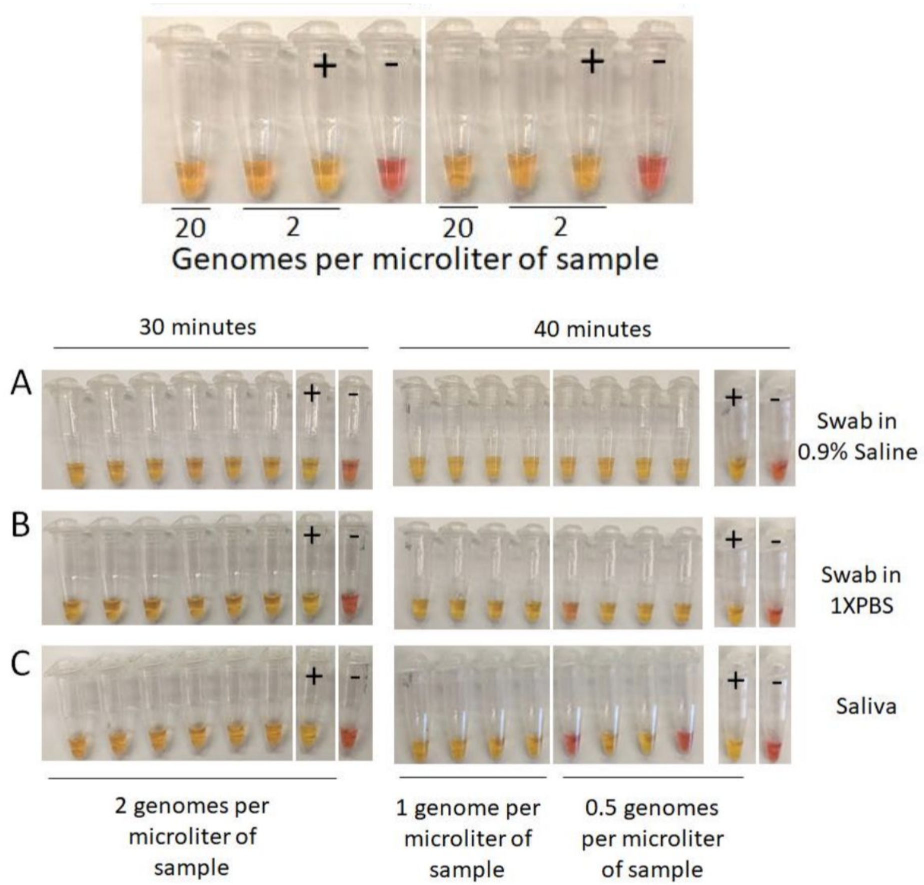


Figure 10. Sensitivity Test HMS Assay 1e Following Sample Purification

Rabe-Cepko Assay Strengths

"Novel sample preparation method that can be used as the first step for many different types of downstream assays."

"Protocol is simple, inexpensive, and rapid."

"Utilizes reagents that are readily prepared in large quantities."

Optimized primers for ORF1ab of SARS-CoV-2: gives very sensitive, high specificity, binary result to at or near gold standard qPCR performance.

Independently chosen by Denali Therapeutics (collaborator to OpenCOVIDScreen)

Execution

Collaborative cooperation of multiple nonprofits, parallel pathing the clinical and regulatory aspects with the practical components and logistics.

Key Execution Strategies

Guiding principle: *speed is crucial* - don't let the perfect be the enemy of the good enough in this pandemic. Any viable solution applied *earlier* is usually far more beneficial.

Recruit one or more clinical partners and proceed to an EUA ASAP.

EUA will be an IVD not LDT, so any CLIA can do it with qualified assay components.

Nonprofit consortium will produce and distribute assay components.

May enable LDT path with key early CLIA lab partners instead, because that allows them independence on reagents. May be faster than getting IVD approval.

Milestones

1st Clinical Partner
v1.0 assay

EUA Submission
to FDA

Pilot Kits
Reagents, Primer
Master Mixes, Controls

International
Collaborations

Expansion
Outreach

1st Priority
Vaccinations

Jun | Jul | Aug | Sept | Oct | Nov | Dec

Mock Pilot
dry run of alpha app

CLIA Cert Extended
able to run real samples

CLIA Pilot

Screening
Schools

Everyone
Screened
at least 1x

U.S. COVID-19
Crisis Effectively
Ended through
Universal Mass
Screening

Thank You!

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Open COVID-19 Mass Screening

June 2020